

Statistics in Health Sciences

Assignment 1

Cluster analysis with missing data using the R packages
mice and miclust

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Instructions

1. It is mandatory to write your assignment integrating R and $\text{\LaTeX}$ in a `.Rnw` document with the `knitr` (as seen in Exercise 0).
2. Respect the numbering sections.
3. For all questions (if not stated otherwise):
 - Write the question
 - Show your R code (i.e., set `echo = TRUE`)
 - Show your R outputs (i.e., set `eval = TRUE`), except for `install.packages()` and ?
 - Answer the question
4. All inline results inserted into your $\text{\LaTeX}$ code must be generated using `\Sexpr{}`.
5. Tables:
 - (a) All tables must be self-contained. I.e., captions must give all details needed to fully understand their contents.
 - (b) You must use `xtable()` (and its companion `print.xtable()`), in the R package `xtable`, for generating $\text{\LaTeX}$ tables in your `.Rnw` documents. **Do not use `knitr::kable()`**. While `kable()` is designed for lightweight Markdown workflows, `xtable()` provides direct, native control over fine-grained $\text{\LaTeX}$ formatting (such as precise column alignment, raw $\text{\LaTeX}$ size control, caption placement, and custom text sanitization), which is required to meet the publication-quality standards of this course. See examples in Exercise 0.
 - (c) Each of those tables must include a caption and a label, which must be defined within the chunk options. See examples in Exercise 0.
 - (d) All tables must be cited in your work (using `\label{}` and `\ref{}`).
6. Figures:
 - (a) All figures must be self-contained. I.e., captions must give all details needed to fully understand their contents.
 - (b) All figures containing results must be created using an R chunk.
 - (c) Each of those figures must include a caption and a label, which must be defined within the chunk options. See examples in Exercise 0.
 - (d) Set the appearance of the figures with the chunk options. For instance, width and height can be controlled with `fig.width` and `fig.height`, respectively (by default, in inches); the figure width relative to the page width can be controlled with `out.width`. Further details can be found at <https://yihui.org/knitr/options/#plots> and Exercise 0.
 - (e) All figures must be cited in your work (using `\label{}` and `\ref{}`).
7. General: Follow all instructions detailed in Section “Trabajos obligatorios en grupo (assignments)” of the document available in “Guía didáctica de la asignatura” on the Campus Virtual.

1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al.^[1] to the data set `copd` in the file `copd_redux.RData`. Data `copd` is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

“The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data.”

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your `pdf` when compiling your `Rnw`. As a hint, you can follow this procedure:
 1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
 2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
 3. To get the thousands comma separator as in “29,070”, you can use `prettyNum()`,
 4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in `LATEX` (as seen in Exercise 0).

2.2 Multiple imputation

Apply multiple imputation to `copd` data using 100 imputations. Explain in a reasoned manner how you decided to set the values of the arguments `method` and `predictorMatrix` when using the `mice()` function. Show your code.

2.3 Integration of multiple imputation in cluster analysis

Reproduce the original analysis in the paper using the `miclust` package to integrate multiple imputation in the cluster analysis. Apply the proposed framework using the same setting than in the original analysis. To do that, follow these steps:

1. Read the help of the `getdata()` function in the `miclust` package. Then, accordingly, create a *list* of *data.frames* and pass it to the `getdata()` function. Give a detailed explanation of the data processing you did to get that *list*. Show your code.
2. Use the `miclust()` function to perform the integration of missing data in cluster analysis using the same settings as in the original analysis. Show your code and explain how and why you set all arguments in the `miclust()` function.

3 Results

Using `plot()` and `summary()` methods in the `miclust` package, provide plots¹ and tables for the results similar than those shown in the original paper. Also, show a barplot for the percentage of times each variable is selected among the imputed data sets, in decreasing order². Explain how you set all arguments in all functions and methods you use. Show your code.

4 Report

Write a brief report (400 words maximum) describing the following issues, and citing tables and figures:

- The final number of clusters and their sizes and uncertainty (cite your figure(s) and table(s)).
- The agreement when relabeling the clusters.
- The final subset of clustering variables and uncertainty.
- The probability of assignment to each cluster and uncertainty.

5 Use of artificial intelligence (AI) technologies

5.1 Instructions

For the completion of this assignment, restricted use of Artificial Intelligence (AI) technologies is permitted. Specifically, the use of AI is allowed exclusively for the following tasks:

- Literature or information search
- Linguistic proofreading and/or translation of text
- Help with L^AT_EX code
- Review of R code that has been previously created by the students without the use of AI

¹Plots in the original paper are in black and white. Here, you can use colors. Hence, explore and use the arguments of the `miclust::plot()` method to provide colored plots. For instance, you can use different colors to identify different metrics. I.e., use the same color in all plots related to the number of clusters, and another color in all plots related to the variable selection.

²For this specific barplot, you can use R code shown in <https://cran.r-project.org/web/packages/miclust/readme/README.html>

1. Clearly and thoroughly identify which contents and results presented in this work have been generated using AI. For each of these contents and results, specify:
 - Identification of the content (e.g., table, figure, R chunk, interpretation, summarization, translation, etc)
 - Name and version of the AI
 - Prompt (question or questions asked to the AI)
2. Write a detailed critical reflection on how the use of the AI tools for this assignment has influenced both the process and the final outcome of the activity.

Lack of transparency regarding the use of AI in this assignment will be considered academic misconduct and may result in a partial or total penalty on the grade for this activity, or more severe sanctions in serious cases.

All authors (i.e., signatories) of this work share equal responsibility for its content and for the methods used to obtain the results.

In general, committing any irregularity in any assessment activity (academic fraud, plagiarism, or improper use of AI, unless explicitly authorized in the course syllabus) that could lead to a significant change in the grade will result in a score of 0 for that assessment. Independent of this, a disciplinary procedure may be initiated against any student committing such irregularities.

5.2 Impact on the assignment score

The content you present in Section 5 will be evaluated in accordance with all details and tasks specified in Section 5.1. The result of this evaluation will be reflected in a score, $I \in [0, 100]$. The value of I will affect the overall score of your assignment as follows. Let A be the score for your assignment (resulting from the remaining sections); the final score for the assignment will be $A \times I/100$. Therefore, values of I less than 100 will act as a penalty on your assignment score.

Delivery:

- **Strictly follow the instructions on delivery of assignments, which are detailed in the “Guía didáctica” of the course.**
- **Deadline: October 28, 2026 at 22:00h.**

References

- [1] X. Basagaña, J. Barrera-Gómez, M. Benet, JM. Antó, and J. Garcia-Aymerich. A Framework for Multiple Imputation in Cluster Analysis. *American Journal of Epidemiology*, 177(7):718–725, 2013. URL <https://doi.org/10.1093/aje/kws289>.